

$x^3+1=u$
 $3x^2 dx=du$
 $\frac{1}{3} \int \frac{1}{u} du$
 $\frac{\ln|u|}{3}$
 $\frac{\ln|x^3+1|}{3}$

$\frac{\sin u}{2}$
 $x^2=u$
 $2x dx=du$
 $\int \frac{\cos u}{2} du$
 $\frac{\sin u}{2}$
 $\frac{\sin x^2}{2}$

$\ln x=u$
 $\frac{1}{x} dx=du$
 $-\cos u$
 $-\cos(\ln x)$

$\ln x=u$
 $\frac{1}{x} dx=du$
 $-\frac{1}{u}$
 $-\frac{1}{\ln x}$

$\ln x=u$
 $\frac{1}{x} dx=du$
 $-\frac{1}{u}$
 $-\frac{1}{\ln x}$

Integral-2

1) $\int \left(\frac{x^2}{x^3+1} + x \cos x^2 + \frac{\sin(\ln x)}{x} - \frac{1}{x \ln^2 x} \right) dx$
 $\frac{\ln|x^3+1|}{3} + \frac{\sin x^2}{2} - \cos(\ln x) + \frac{1}{\ln x} + c$

2) $\int \left(\frac{x}{x^2+1} - \cos x e^{2+\sin x} + \frac{4 \tan x}{\cos^2 x} \right) dx$
 $\frac{\ln|x^2+1|}{2} - e^{2+\sin x} + \frac{4 \tan x}{\ln 4} + c$

3) $\int \left(\frac{e^x}{e^x+1} + \frac{\cos x}{1+\sin^2 x} + \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} \right) dx$
 $\ln|e^x+1| + \arctan(\sin x) + \arcsin(\tan x) + c$

Bu integralleri
 değişken
 dönüşümünü
 uzun uzun
 yazarak
 değil, kime
 u diyeceğinizi
 ve ortaya
 çıkacak integrali
 hızlıca düşünerek
 yazarak çözmeye
 çalışın.

Örneğin: $\int \frac{\sec^2 x}{1+\tan^2 x} dx = \arctan(\tan x) + c$

$\tan^2 x = u$

$\int \frac{du}{1+u} \rightarrow \arctan u$

$\int \tan x \cdot \sec^2 x dx = 2u$ gibi hızlı çözüm yapma-
 yı deneyin 😊

$\frac{-2}{\sqrt{7}\sqrt{1-\frac{x^2}{7}}}$
 $\frac{-2}{\sqrt{7}} \cdot \frac{1}{\sqrt{1-\frac{x^2}{7}}}$
 $\int \frac{1}{1+u^2} = \arctan u$
 $-\arctan(\tan x) + c$

4) $\int \left(\frac{3}{x^2+6} - \frac{2}{\sqrt{7-x^2}} + \cos^2 x \right) dx = ?$

$\frac{3}{x^2+6}$
 $\frac{3}{\sqrt{6}} \arctan \frac{x}{\sqrt{6}}$
 $\frac{2}{\sqrt{7-x^2}}$
 $2 \arcsin \frac{x}{\sqrt{7}}$
 $\cos^2 x = 2\cos^2 x - 1$
 $\frac{\cos 2x}{2}$
 $\frac{\sin 2x}{4} + \frac{x}{2} + c$
 $\int \frac{1}{1+\frac{x^2}{6}} = \frac{3}{\sqrt{6}} \arctan \frac{x}{\sqrt{6}}$

$$x^2 = u \quad 2x dx = du \quad x^2 e^x = \int e^x 2x dx \quad 2x = u \quad 2 dx = du \quad 2x e^x = \int e^x 2x dx$$

$$\textcircled{5} \int_0^1 x^2 e^x dx = ? \quad x^2 e^x - 2x e^x + 2e^x \Big|_0^1 = e - 2e + 2e = e - 2$$

$$\textcircled{6} \int x \cos x dx = ? \quad x \sin x + \cos x$$

$$\textcircled{7} \int x \ln x dx = ? \quad \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

$$\textcircled{8} \int x \cdot \text{Arctan} x dx = ? \quad \frac{x^2}{2} \text{Arctan} x - \frac{x^2}{4} \text{Arctan} x + C$$

$$\textcircled{9} \int (x^2 - 5x) e^x dx = ? \quad x^2 - 5x \cdot e^x - \int e^x 2x - 5 dx = x^2 - 5x \cdot e^x - 2x - 5e^x + 2e^x + C$$

$$\textcircled{10} \int \cos^5 x dx = ? \quad \frac{\sin x - 2 \sin^3 x + \sin^5 x}{3} + C$$

$$\textcircled{11} \int \cos^3 x \cdot \sin^7 x dx = ? \quad \frac{\sin^8 x}{8} - \frac{\sin^{10} x}{10} + C$$

$$\textcircled{12} \int \cos^8 x \cdot \sin^3 x dx = ? \quad \frac{-\cos^9 x}{9} + \frac{\cos^{11} x}{11} + C$$

$$\textcircled{13} \int \sec^5 x \cdot \tan^5 x dx = ? \quad \frac{\sec^2 x - 2 \sec^4 x + \sec^6 x}{5} + C$$

$$\textcircled{14} \int \sec^4 x \cdot \tan^5 x dx = ? \quad \frac{\tan^6 x}{6} + \frac{\tan^8 x}{8} + C$$

$$\textcircled{15} \int \sec^6 x dx = ? \quad \tan x + \frac{2 \tan^3 x}{3} + \frac{\tan^5 x}{5} + C$$

Kısmi integrasyon soruları. Bu sorulardan uygun olanları hem klasik yöntemle hem de tablo yöntemiyle çözmeyi deneyin olur mu.

$x^2 - 5x$	e^x
$2x - 5$	e^x
2	e^x
0	e^x

$$2x - 5 = u \quad 2 dx = du \quad 2x - 5 \cdot e^x - \int e^x 2x - 5 dx$$

$$e^x dx = du \quad e^x = v$$

$$2x - 5 \cdot e^x - 2e^x$$

16) $\int \frac{x^2}{\sqrt{9-x^2}} dx = ?$

Trigonometric substitution: $x = 3 \sin t$, $dx = 3 \cos t dt$, $\sqrt{9-x^2} = 3 \cos t$

Integral becomes: $\int \frac{9 \sin^2 t \cdot 3 \cos t}{3 \cos t} dt = 9 \int \sin^2 t dt$

Using identity $\sin^2 t = \frac{1 - \cos 2t}{2}$:

$9 \int \frac{1 - \cos 2t}{2} dt = \frac{9}{2} \left(t - \frac{\sin 2t}{2} \right) + C$

Back substitution: $t = \arcsin \frac{x}{3}$, $\sin 2t = 2 \sin t \cos t = \frac{2x}{3} \cdot \frac{\sqrt{9-x^2}}{3} = \frac{2x\sqrt{9-x^2}}{9}$

Final answer: $\frac{9}{2} \left(\arcsin \frac{x}{3} - \frac{2x\sqrt{9-x^2}}{9} \right) + C$

17) $\int \frac{dx}{\sqrt{25x^2-4}}$

Trigonometric substitution: $5x = \frac{2}{\sec t}$, $dx = -\frac{2}{\sec^2 t} \tan t dt$, $\sqrt{25x^2-4} = \frac{2 \tan t}{\sec t} = 2 \sin t$

Integral becomes: $\int \frac{-2 \tan t}{2 \sin t} dt = -\int \frac{1}{\cos t} dt = -\ln |\sec t + \tan t| + C$

Back substitution: $\sec t = \frac{5x}{2}$, $\tan t = \frac{\sqrt{25x^2-4}}{2}$

Final answer: $-\ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2} \right| + C$

18) $\int \frac{x^3 dx}{\sqrt{4x^2+9}}$

Trigonometric substitution: $2x = 3 \tan t$, $dx = 3 \sec^2 t dt$, $\sqrt{4x^2+9} = 3 \sec t$

Integral becomes: $\int \frac{27 \tan^3 t \cdot 3 \sec^2 t}{27 \sec^3 t} dt = \int \tan^2 t dt$

Using identity $\tan^2 t = \sec^2 t - 1$:

$\int (\sec^2 t - 1) dt = \tan t - t + C$

Back substitution: $\tan t = \frac{2x}{3}$, $t = \arctan \frac{2x}{3}$

Final answer: $\frac{2x}{3} - \arctan \frac{2x}{3} + C$

19) $\int \frac{dx}{(x+1)^3 \sqrt{x^2+2x-3}}$

Trigonometric substitution: $x+1 = \frac{4}{\sec t}$, $dx = -\frac{4}{\sec^2 t} \tan t dt$, $\sqrt{x^2+2x-3} = \frac{4 \tan t}{\sec t} = 4 \sin t$

Integral becomes: $\int \frac{-4 \tan t}{64 \sin^3 t} dt = -\frac{1}{16} \int \frac{1}{\sin^2 t} dt = -\frac{1}{16} \int \csc^2 t dt = \frac{1}{16} \cot t + C$

Back substitution: $\cot t = \frac{\cos t}{\sin t} = \frac{\frac{4}{\sec t}}{4 \sin t} = \frac{1}{\sec t \sin t} = \frac{\cos t}{2}$

Final answer: $\frac{1}{16} \cot t + C$

20) $\int \frac{x^2 dx}{\sqrt{4+x^2}}$

Trigonometric substitution: $x = 2 \tan t$, $dx = 2 \sec^2 t dt$, $\sqrt{4+x^2} = 2 \sec t$

Integral becomes: $\int \frac{4 \tan^2 t \cdot 2 \sec^2 t}{4 \sec^3 t} dt = \int \tan^2 t dt$

Using identity $\tan^2 t = \sec^2 t - 1$:

$\int (\sec^2 t - 1) dt = \tan t - t + C$

Back substitution: $\tan t = \frac{x}{2}$, $t = \arctan \frac{x}{2}$

Final answer: $\frac{x}{2} - \arctan \frac{x}{2} + C$

21) $\int \sqrt{5-3x^2} dx$

Trigonometric substitution: $\sqrt{5} \cos t = \sqrt{5-3x^2}$, $dx = -\frac{\sqrt{5}}{3} \sin t dt$

Integral becomes: $\int \sqrt{5} \cos t \cdot -\frac{\sqrt{5}}{3} \sin t dt = -\frac{5}{3} \int \cos t \sin t dt = -\frac{5}{6} \int \sin 2t dt = \frac{5}{12} \cos 2t + C$

Back substitution: $\cos 2t = 2 \cos^2 t - 1 = 2 \left(\frac{\sqrt{5-3x^2}}{\sqrt{5}} \right)^2 - 1 = \frac{2(5-3x^2)}{5} - 1 = \frac{10-6x^2-5}{5} = \frac{5-6x^2}{5}$

Final answer: $\frac{5}{12} \left(\frac{5-6x^2}{5} \right) + C = \frac{5-6x^2}{12} + C$

22) $\int \sqrt{5-3x^2} dx$

Trigonometric substitution: $\sqrt{5} \cos t = \sqrt{5-3x^2}$, $dx = -\frac{\sqrt{5}}{3} \sin t dt$

Integral becomes: $\int \sqrt{5} \cos t \cdot -\frac{\sqrt{5}}{3} \sin t dt = -\frac{5}{3} \int \cos t \sin t dt = -\frac{5}{6} \int \sin 2t dt = \frac{5}{12} \cos 2t + C$

Back substitution: $\cos 2t = 2 \cos^2 t - 1 = 2 \left(\frac{\sqrt{5-3x^2}}{\sqrt{5}} \right)^2 - 1 = \frac{2(5-3x^2)}{5} - 1 = \frac{10-6x^2-5}{5} = \frac{5-6x^2}{5}$

Final answer: $\frac{5}{12} \left(\frac{5-6x^2}{5} \right) + C = \frac{5-6x^2}{12} + C$

23) $\int \sqrt{5-3x^2} dx$

Trigonometric substitution: $\sqrt{5} \cos t = \sqrt{5-3x^2}$, $dx = -\frac{\sqrt{5}}{3} \sin t dt$

Integral becomes: $\int \sqrt{5} \cos t \cdot -\frac{\sqrt{5}}{3} \sin t dt = -\frac{5}{3} \int \cos t \sin t dt = -\frac{5}{6} \int \sin 2t dt = \frac{5}{12} \cos 2t + C$

Back substitution: $\cos 2t = 2 \cos^2 t - 1 = 2 \left(\frac{\sqrt{5-3x^2}}{\sqrt{5}} \right)^2 - 1 = \frac{2(5-3x^2)}{5} - 1 = \frac{10-6x^2-5}{5} = \frac{5-6x^2}{5}$

Final answer: $\frac{5}{12} \left(\frac{5-6x^2}{5} \right) + C = \frac{5-6x^2}{12} + C$

24) $\int \sqrt{5-3x^2} dx$

Trigonometric substitution: $\sqrt{5} \cos t = \sqrt{5-3x^2}$, $dx = -\frac{\sqrt{5}}{3} \sin t dt$

Integral becomes: $\int \sqrt{5} \cos t \cdot -\frac{\sqrt{5}}{3} \sin t dt = -\frac{5}{3} \int \cos t \sin t dt = -\frac{5}{6} \int \sin 2t dt = \frac{5}{12} \cos 2t + C$

Back substitution: $\cos 2t = 2 \cos^2 t - 1 = 2 \left(\frac{\sqrt{5-3x^2}}{\sqrt{5}} \right)^2 - 1 = \frac{2(5-3x^2)}{5} - 1 = \frac{10-6x^2-5}{5} = \frac{5-6x^2}{5}$

Final answer: $\frac{5}{12} \left(\frac{5-6x^2}{5} \right) + C = \frac{5-6x^2}{12} + C$

Cevaplar

$$\textcircled{1} \int \left(\frac{x^2}{x^3+1} + x \cos x^2 + \frac{\sin(\ln x)}{x} - \frac{1}{x \ln^2 x} \right) dx$$

$$= \int \underbrace{\frac{x^2 dx}{x^3+1}}_{\substack{\text{ici: } \frac{du}{3u} \\ \downarrow \\ \text{integrali: } \frac{1}{3} \ln u \\ \downarrow u=x^3+1}} + \int \underbrace{\cos x^2 \cdot x dx}_{\substack{\text{ici: } \frac{\cos u du}{2} \\ \downarrow \\ \text{int: } \frac{\sin u}{2} \\ \downarrow u=x^2}} + \int \underbrace{\frac{\sin(\ln x)}{x} dx}_{\substack{\text{ici: } \sin u du \\ \downarrow \\ \text{int: } -\cos u \\ \downarrow u=\ln x}} - \int \underbrace{\frac{1}{\ln^2 x} \cdot \frac{dx}{x}}_{\substack{\text{ici: } \frac{1}{u^2} du \\ \downarrow \\ \text{int: } -\frac{1}{u} \\ \downarrow u=\ln x}}$$

$$= \frac{1}{3} \ln|x^3+1| + \frac{1}{2} \sin x^2 - \cos(\ln x) + \frac{1}{\ln x} + C$$

$$\textcircled{2} \int \left(\frac{x}{x^2+1} - \cos x \cdot e^{2+\sin x} + \frac{4^{\tan x}}{\cos^2 x} \right) dx$$

$$= \int \underbrace{\frac{x dx}{x^2+1}}_{\substack{\text{ici: } \frac{du}{2u} \\ \downarrow \\ \text{integrali: } \frac{\ln u}{2} \\ \downarrow u=x^2+1}} - \int \underbrace{e^{2+\sin x} \cdot \cos x dx}_{\substack{\text{ici: } e^u du \\ \downarrow \\ \text{int: } e^u \\ \downarrow u=2+\sin x}} + \int \underbrace{4^{\tan x} \cdot \sec^2 x dx}_{\substack{\text{ici: } 4^u du \\ \downarrow \\ \text{int: } \frac{4^u}{\ln 4} \\ \downarrow u=\tan x}}$$

$$= \frac{1}{2} \ln|x^2+1| - e^{2+\sin x} + \frac{4^{\tan x}}{\ln 4} + C$$

$$\textcircled{3} \int \left(\frac{e^x}{e^x+1} + \frac{\cos x}{1+\sin^2 x} + \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} \right) dx$$

$$= \int \frac{\overbrace{e^x dx}^{du}}{\underbrace{e^x+1}_u} + \int \frac{\overbrace{\cos x dx}^{du}}{1+\underbrace{\sin^2 x}_{u^2}} + \int \frac{\overbrace{\sec^2 x dx}^{du}}{\sqrt{1-\underbrace{\tan^2 x}_{u^2}}}$$

$$\downarrow u=e^x+1$$

$$\downarrow u=\sin x$$

$$\downarrow u=\tan x$$

$$= \int \frac{du}{u} + \int \frac{du}{1+u^2} + \int \frac{du}{\sqrt{1-u^2}}$$

$$\downarrow$$

$$\downarrow$$

$$= \ln|u| + \text{Arctan } u + \text{Arcsin } u + C$$

$$\downarrow u=e^x+1$$

$$\downarrow u=\sin x$$

$$\downarrow u=\tan x$$

$$= \ln|e^x+1| + \text{Arctan}(\sin x) + \text{Arcsin}(\tan x) + C$$

$$\textcircled{4} \int \left(\frac{3}{x^2+6} - \frac{2}{\sqrt{7-x^2}} + \frac{\cos^2 x}{1+\cos 2x} \right) dx = ?$$

$$\frac{3}{6} \cdot \frac{1}{1+\frac{x^2}{6}} \quad \frac{1}{2} \cdot \frac{1}{\frac{1}{2}\left(\frac{x}{\sqrt{6}}\right)^2}$$

$$= \frac{3}{\sqrt{6}} \cdot \text{Arctan} \frac{x}{\sqrt{6}} - 2 \cdot \text{Arcsin} \frac{x}{\sqrt{7}} + \frac{x}{2} + \frac{\sin 2x}{4} + C$$

$$\frac{\sqrt{6}}{2} \cdot \text{Arctan} \frac{x}{\sqrt{6}}$$

$$\frac{6}{2\sqrt{6}} \quad \frac{3}{\sqrt{6}} \text{ arctan } \frac{x}{\sqrt{6}}$$

KURAL

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a} \text{Arctan} \frac{x}{a}$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \text{Arcsin} \frac{x}{a}$$

$$\textcircled{5} \int_0^1 x^2 e^x dx = ?$$

I.401 $I = \int_0^1 x^2 e^x dx$

LAP T Ü

1. Kısmi int. $\begin{cases} u = x^2 \rightarrow du = 2x dx \\ dv = e^x dx \rightarrow v = e^x \end{cases}$

$$\int u dv = u \cdot v - \int v du$$

$$I = x^2 e^x - 2 \int_0^1 x e^x dx \rightarrow 2. \text{ Kısmi gerekiyor}$$

$$= x^2 e^x - 2 [x e^x - \int e^x dx]$$

$$\begin{cases} u = x \rightarrow du = dx \\ dv = e^x dx \rightarrow v = e^x \end{cases}$$

$$= x^2 e^x - 2x e^x + 2e^x$$

$$\int_0^1 x^2 e^x dx = (x^2 e^x - 2x e^x + 2e^x) \Big|_0^1 =$$

$$= e - 2e + 2e - 2 = \boxed{e-2}$$

II.401 : Tablo

$$I = \int x^2 e^x dx$$

LAP T Ü

Türev		İntegral
x^2	$+$	e^x
$\downarrow$		$\downarrow$
$2x$	$-$	e^x
$\downarrow$		$\downarrow$
2	$+$	e^x
$\downarrow$		$\downarrow$
0		e^x

$$I = x^2 e^x - 2x e^x + 2e^x$$

$$\int_0^1 x^2 e^x dx = (x^2 e^x - 2x e^x + 2e^x) \Big|_0^1 = e - 2$$

$$⑥ \int \underbrace{x}_P \underbrace{\cos x}_T dx = ?$$

LAPTÜ

I. Yol:

$$u = x \xrightarrow{T} du = dx$$

$$dv = \cos x dx \xrightarrow{I} v = \sin x$$

$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C$$

II. Yol: Tablo

<u>Türev</u>		<u>Integral</u>
x	+	cos x
1	-	sin x
0		-cos x

$$\left. \begin{array}{l} \text{cos x} \\ \text{sin x} \\ \text{-cos x} \end{array} \right\} I = x \sin x + \cos x + C$$

$$⑦ \int \underbrace{x}_P \underbrace{\ln x}_L dx = ?$$

LAPTÜ

★★
→ Tablo yöntemi
olmaz. $\ln x$ e v
diyeceğiz ★★

$$u = \ln x \xrightarrow{T} du = \frac{1}{x} dx$$

$$dv = x dx \xrightarrow{I} v = \frac{x^2}{2}$$

$$\int x \ln x dx = \frac{x^2}{2} \cdot \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

LAPTÜ

$$\textcircled{8} \int \underbrace{x}_{\text{P}} \cdot \underbrace{\text{Arctan} x}_{\text{A}} dx = ?$$

LAP T $\bar{U}$ $\rightarrow$ Tablo olmaz!

$$u = \text{Arctan} x \xrightarrow{\text{T}} du = \frac{1}{1+x^2} dx$$

$$dv = x dx \xrightarrow{\text{I}} v = \frac{x^2}{2}$$

$$\int x \text{Arctan} x dx = \frac{x^2}{2} \cdot \text{Arctan} x - \frac{1}{2} \int \overbrace{\frac{x^2}{1+x^2}}^{x^2+1-1} dx$$

$x^2+1-1 \rightarrow$ Paya 1 ekleyip
1 çıkaralım

$$= \frac{x^2}{2} \text{Arctan} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2} \right) dx$$

$\downarrow \quad \downarrow$

$$= \frac{x^2}{2} \text{Arctan} x - \frac{1}{2} (x - \text{Arctan} x) + c$$

$$\textcircled{9} \int \overbrace{(x^2 - 5x)}^{\text{P}} \overbrace{e^x}^{\bar{U}} dx = ? \quad \text{LAP T}\bar{U}$$

I.401

$$u = x^2 - 5x \rightarrow du = (2x - 5) dx$$

$$dv = e^x dx \rightarrow v = e^x$$

$$u = 2x - 5 \rightarrow dv = 2 dx$$

$$I = \overbrace{(x^2 - 5x)}^{\text{P}} \overbrace{e^x}^{\bar{U}} - \int \overbrace{(2x - 5)}^{\text{P}} \overbrace{e^x}^{\bar{U}} dx \rightarrow dv = e^x dx \rightarrow v = e^x$$

$$= (x^2 - 5x) e^x - \left[(2x - 5) \cdot e^x - \int 2e^x dx \right]$$

$$= (x^2 - 5x) e^x - (2x - 5) e^x + 2e^x + c$$

II.401 Tablo

<u>Türev</u>		<u>integral</u>
$x^2 - 5x$		e^x
↓	+	↓
$2x - 5$		e^x
↓	-	↓
2		e^x
↓	+	↓
0		e^x

$$I = (x^2 - 5x)e^x - (2x - 5)e^x + 2e^x + C$$

⑩ $\int \cos^5 x \, dx = ?$ $\cos^5 x \cdot (\sin x)^0 \rightarrow$ Üstlerin biri tek biri çift, çift olanı u demeliyiz.

$$\int \underbrace{\cos^4 x}_{(\cos^2 x)^2} \cdot \cos x \, dx = \int (1 - \sin^2 x)^2 \cdot \cos x \, dx$$

$$= \int (1 - u^2)^2 \, du = \int (1 - 2u^2 + u^4) \, du$$

$$u = \sin x$$

$$= u - \frac{2}{3} u^3 + \frac{u^5}{5} + C$$

$$du = \cos x \, dx$$

$$= \sin x - \frac{2}{3} \sin^3 x + \frac{\sin^5 x}{5} + C$$

⑪ $\int \cos^3 x \cdot \sin^7 x \, dx = ? \rightarrow$ ikisi de tek, iki dönüşüm de olur.

I.401 $u = \sin x \quad du = \cos x \, dx$

$$\int \underbrace{\cos^2 x}_{1-u^2} \cdot \underbrace{\sin^7 x}_{u^7} \cdot \underbrace{\cos x \, dx}_{du} = \int (u^7 - u^9) \, du = \frac{(\sin x)^8}{8} - \frac{(\sin x)^{10}}{10} + C$$

II.401 $u = \cos x \rightarrow du = -\sin x dx$

$$\int \underbrace{\cos^3 x}_{u^3} \cdot \underbrace{(\sin^2 x)^3}_{(1-u^2)^3} \cdot \underbrace{\sin x dx}_{-du} \Rightarrow \text{Bunu da siz sonuclandırın 😊}$$

12) $\int \cos^8 x \cdot \sin^3 x dx = ? \rightarrow \star \cos x \text{ in üstü çift } \star$

$u = \cos x \quad du = -\sin x dx$

$$\begin{aligned} \int \underbrace{\cos^8 x}_{u^8} \cdot \underbrace{\sin^2 x}_{1-u^2} \cdot \underbrace{\sin x dx}_{-du} &= - \int (u^8 - u^{10}) du \\ &= - \frac{(\cos x)^9}{9} + \frac{(\cos x)^{11}}{11} + C \end{aligned}$$

13) $\int \sec^5 x \cdot \tan^3 x dx = \int \underbrace{\sec^4 x}_{u^4} \cdot \underbrace{\tan^2 x}_{(u^2-1)^2} \cdot \underbrace{\sec x \cdot \tan x dx}_{du}$

$u = \sec x$

$du = \sec x \tan x dx$

$\tan^2 x = \sec^2 x - 1$

$$= \int u^4 \cdot (u^2 - 1)^2 du$$

$$= \int (u^8 - 2u^6 + u^4) du$$

$$= \frac{(\sec x)^9}{9} - 2 \cdot \frac{(\sec x)^7}{7} + \frac{(\sec x)^5}{5} + C$$

$$(14) \int \sec^4 x \cdot \tan^5 x \, dx = ?$$

I. 401

$$\int \underbrace{\sec^2 x}_{u^2+1} \cdot \underbrace{\tan^5 x}_{u^5} \cdot \underbrace{\sec^2 x \, dx}_{du} = \int (u^7 + u^5) \, du$$

$$= \frac{(\tan x)^8}{8} + \frac{(\tan x)^6}{6} + C$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$1 + \tan^2 x = \sec^2 x$$

II. 401

$$\int \underbrace{\sec^3 x}_{u^3} \cdot \underbrace{(\tan^2 x)^2}_{(u^2-1)^2} \cdot \underbrace{\sec x \cdot \tan x \, dx}_{du}$$

$$u = \sec x$$

$$du = \sec x \cdot \tan x \, dx$$

$$\tan^2 x = \sec^2 x - 1$$

$$= \int u^3 \cdot (u^4 - 2u^2 + 1) \, du$$

$$= \int (u^7 - 2u^5 + u^3) \, du$$

$$= \frac{(\sec x)^8}{8} - 2 \cdot \frac{(\sec x)^6}{6} + \frac{(\sec x)^4}{4} + C$$

$$(15) \int \sec^6 x \, dx = \int \underbrace{(\sec^2 x)^2}_{(u^2+1)^2} \cdot \underbrace{\sec^2 x \, dx}_{du} = \int (u^4 + 2u^2 + 1) \, du$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$\sec^2 x = \tan^2 x + 1$$

$$= \frac{(\tan x)^5}{5} + 2 \frac{(\tan x)^3}{3} + \tan x + C$$

$$(16) \int \frac{x^2}{\sqrt{9-x^2}} dx = ?$$

$$x = 3 \sin t$$

$$dx = 3 \cos t dt$$

$$\sqrt{9-x^2} = \sqrt{9-9\sin^2 t} = 3 \cos t$$

Planmiz



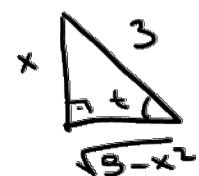
$$\int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9 \sin^2 t \cdot 3 \cos t dt}{3 \cos t} = \int 9 \sin^2 t dt$$

$$= 9 \int \frac{1 - \cos 2t}{2} dt = \frac{9}{2} \left[t - \frac{\sin 2t}{2} \right] + c$$

$$= \frac{9}{2} \left[t - \cos t \cdot \sin t \right] + c$$

$$= \frac{9}{2} \left[\arcsin \frac{x}{3} - \frac{\sqrt{9-x^2}}{3} \cdot \frac{x}{3} \right] + c$$

$$\frac{x}{3} = \sin t$$



$$(17) \int \frac{dx}{\sqrt{25x^2-4}} = ?$$

$$5x = 2 \sec t$$

$$5dx = 2 \sec t \tan t dt$$

$$\sqrt{25x^2-4} = \sqrt{4\sec^2 t - 4} = 2 \tan t$$

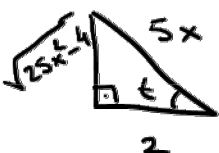
Plan

$$\int \frac{dx}{\sqrt{25x^2-4}} = \int \frac{\frac{2}{5} \sec t \tan t dt}{2 \tan t} = \frac{1}{5} \int \sec t dt$$

$$\sec t = \frac{5x}{2}$$

$$= \frac{1}{5} \ln |\sec t + \tan t| + c$$

$$= \frac{1}{5} \ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2} \right| + c$$



18) $\int \frac{x^3 dx}{\sqrt{4x^2+9}}$ int. cözmek için aşağıdaki dönüşümlerden yapılmalıdır?

- a) $x=3\tan t$ **b) $2x=3\tan t$** c) $2x=3\sec t$ d) $2x=3\sin t$

19) $\int \frac{dx}{(x+1)^3 \sqrt{x^2+2x-3}}$ ($x>1$) int. cözmek için uygun bir trigonometrik dönüşüm yapıldığında aşağıdaki int. hangisi elde edilir?

- a) $\frac{1}{8} \int \sin^2 t dt$ **b) $\frac{1}{8} \int \cos^2 t dt$** c) $\frac{1}{4} \int \sec^2 t dt$ d) $\frac{1}{4} \int \tan^2 t dt$

$$x^2+2x-3 = (x+1)^2-4$$

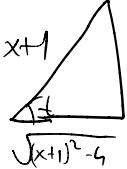
$$\int \frac{dx}{(x+1)^3 \sqrt{(x+1)^2-4}}$$

$(2)^2$

$$x+1 = 2\sec t$$

$$dx = 2\sec t \tan t dt$$

$$\sqrt{(x+1)^2-4} = \sqrt{4\sec^2 t-4} = 2\tan t$$



$\sin t = \frac{x+1}{2}$
 $2\csc t = x+1$
 $-2\csc t \cdot \cot t \cdot dt = dx$
 $\int \frac{-2\csc t \cdot \cot t \cdot dt}{8\csc^3 t \cdot 2\cot t}$
 $= -\frac{1}{8} \int \sin^2 t$

$$= \int \frac{2\cancel{\sec t} \cancel{\tan t} dt}{8 \cdot \cancel{\sec^3 t} \cdot 2\cancel{\tan t}} = \frac{1}{8} \int \frac{1}{\sec^2 t} dt = \frac{1}{8} \int \cos^2 t dt$$

$\sec^2 t$

20) $\int \frac{x^2 dx}{\sqrt{4+x^2}}$ integralinin uygun bir trigonometrik dönüşüm altında alacağı yeni hal aşağıdakilerden hangisi olabilir?

a) $\int 4 \tan^2 \theta \cdot \sec \theta d\theta$ b) $\int 2 \tan^2 \theta \cos \theta d\theta$ c) $\int 8 \tan^2 \theta \cdot \sec^2 \theta d\theta$

d) $\int 4 \tan^2 \theta \cdot \cos \theta d\theta$ e) $\int 4 \sec^3 \theta d\theta$

$$\left. \begin{array}{l} x = 2 \tan t \\ dx = 2 \sec^2 t dt \\ \sqrt{4+x^2} = \sqrt{4+4\tan^2 t} = 2 \sec t \end{array} \right\} \text{Plan}$$

$$\int \frac{x^2 dx}{\sqrt{4+x^2}} = \int \frac{4 \cdot \tan^2 t \cdot \cancel{2 \sec^2 t} dt}{\cancel{2 \sec t}} = \int 4 \tan^2 t \cdot \sec t dt$$

21) $\int \sqrt{5-3x^2} dx$ int. cozmek için uygun bir trigonometrik dönüşüm yapıldığında aşağıdaki int. hangisi elde edilir?

a) $\int \frac{5}{3} \cos^2 t dt$ b) $\int \frac{\sqrt{5}}{\sqrt{3}} \sec^2 t dt$ c) $\int \frac{5}{\sqrt{3}} \cos^2 t dt$

$$\sqrt{3}x = \sqrt{5} \sin t \rightarrow x = \frac{\sqrt{5}}{\sqrt{3}} \sin t$$

$$dx = \frac{\sqrt{5}}{\sqrt{3}} \cos t dt$$

$$\sqrt{5-3x^2} = \sqrt{5-5\sin^2 t} = \sqrt{5} \cos t$$

$$\int \sqrt{5-3x^2} dx = \int \sqrt{5} \cos t \cdot \frac{\sqrt{5}}{\sqrt{3}} \cos t dt = \frac{5}{\sqrt{3}} \int \cos^2 t dt$$